



SIPCOT SIMS

Smart Industrial Monitoring System User Guide & Operating Manual

Admin Portal

Government Dashboard

Industry Workspace

Version 2.0 | April 2026

State Industries Promotion Corporation of Tamil Nadu

Government of Tamil Nadu

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1. System Overview

The SIPCOT Smart Industrial Monitoring System (SIMS) is a data-driven Industrial Operating System designed to transform Tamil Nadu's industrial governance. It provides real-time monitoring, compliance tracking, and analytics for industrial parks across the state.

Three User Roles

Admin

Full system control, user management, compliance oversight, park management, and system configuration.

Government Officer

State-wide monitoring, command center, compliance engine access, analytics dashboards, and report generation.

Industry User

Personal workspace, data submission, compliance score tracking, document management, and service requests.

Public Visitor

Home page, industrial parks overview, about page, features, and contact support - no login required.

Technology Stack

Component	Technology
Frontend	React 19 + Material UI 7 + Recharts
Backend	Express 5 + Node.js
Database	PostgreSQL
Authentication	JWT (JSON Web Tokens) with RBAC

Build Tool

Vite 8

2. Getting Started

How to Access the Portal

1 Open the Application

Navigate to the home page URL in your browser. The public landing page will load showing Tamil Nadu's industrial pulse data.

2 Click "Portal Login"

In the top navigation bar, click the "Portal Login" button. This takes you to the Role Selection page.

3 Select Your Role

Choose from three portals: **Admin Portal**, **Industry Portal**, or **Government Portal**. Each has a distinct color and icon.

4 Enter Credentials & Sign In

Enter your email and password. The system validates your role matches the selected portal. On success, you're redirected to your role-specific dashboard.

Tip: Use the "Forgot Password?" link if you need to reset your credentials. The system will send instructions to your registered email.

3. Demo Login Credentials

Use the following test credentials to explore each role:

Admin Portal

Full system access

Email: admin@sipcot.com

Password: (any value - mock mode)

Redirects to: Admin Dashboard (/admin-dashboard)

Government Officer Portal

State-wide monitoring & oversight

Email: govt@tn.gov.in

Password: (any value - mock mode)

Redirects to: Command Center (/command-center)

Industry Portal

Company workspace & compliance

Email: industry@abc.com

Password: (any value - mock mode)

Redirects to: Industry Workspace (/workspace)

Note: In the current demo mode, any password is accepted. In production, bcrypt-hashed passwords will be validated against the PostgreSQL database.

4. Admin Portal Guide

Login: admin@sipcot.com

Sidebar Navigation

OVERVIEW

- Admin Dashboard - KPI overview with charts
- Command Center - State-wide industrial monitoring

MANAGEMENT

- Parks Explorer - All 8 industrial parks
- Services Tracker - Kanban-style service request tracking
- User Management - Add/edit/approve users

ANALYTICS & COMPLIANCE

- Compliance Engine - Violation tracking & predictions
- Analytics - Investment & employment charts
- Report Center - Generate PDF/Excel/CSV reports

SYSTEM

- Settings - Security, backup, system config

Admin Dashboard /admin-dashboard

The main landing page after admin login. Shows:

- **4 KPI Cards:** Total Industries, Total Investment, Employment, Power Usage
- **Industry Growth Chart:** Line chart showing 5-year growth (2021-2025)
- **Resource Consumption:** Bar chart comparing resource usage by park

Command Center /command-center

High-level state-wide monitoring dashboard:

- **5 KPI Cards:** Total CapEx (Rs. 24,500 Cr), Revenue, Direct Jobs, Indirect Jobs, Red Flags - each with growth percentage and trend indicators
- **District Investment Chart:** Horizontal bar chart showing investment distribution across 5 districts
- **Park Rankings Table:** Sortable table ranking parks by infrastructure score, industries, or investment. Top 3 parks highlighted in blue
- **Alerts Panel:** Color-coded alerts (critical=red, high=orange, medium=yellow, low=blue) with action counts
- **5-Year Growth Trend:** Toggle between Investment and Employment trend lines
- **Activity Feed:** Real-time log of system events (submissions, violations, registrations)

Services Tracker /services

Kanban-style lifecycle tracker for NOCs, allotments, and service requests:

- **Summary Cards:** Applied, In Review, Pending Approval, Completed counts
- **View Toggle:** Switch between Kanban board and Table view
- **Kanban Columns:** Applied > Doc Review > Inspection > Pending Approval > Approved > Completed
- **Click any card:** Opens detail dialog with milestone timeline (stepper), bottleneck alerts, and action buttons
- **Update Status button:** Admin/Govt can update service request status with confirmation

User Management /user-management

Full CRUD for system users:

- **User Table:** Name, Email, Role (color-coded chip), Status, Actions
- **" + Add User" button:** Opens dialog to create new user with name, email, role, status fields
- **Edit (pencil icon):** Opens pre-filled edit dialog
- **Approve (check icon):** Activates pending users
- **Delete (trash icon):** Removes user with confirmation

Compliance Engine /compliance-engine

Automated compliance monitoring and analytics:

- **Overview Cards:** Compliant (892), Warning (156), Violations (45), Missing Data (67)
- **Three Tabs:** Violations | Trends & Analytics | Predictions
- **Violations Tab:** Filterable table with Severity/Status dropdowns. Each row has "Acknowledge" and "Escalate" buttons that update status with confirmation
- **Missing Submissions Alert:** Shows count + deadline. "Send Bulk Reminder" sends email notifications. "Export List" downloads CSV file
- **Trends Tab:** 9-month compliance score line chart
- **Predictions Tab:** Table showing current vs. projected (1yr) for Investment, Employment, Compliance Rate, Active Industries. "Download Full Report" exports CSV

Report Center /report-center

Self-service report generation:

- **Step 1 - Report Type:** Select Investment Summary, Employment, Compliance, Park Performance, Resource Utilization, or Custom Query
- **Step 2 - Filters:** Park, Period, Quarter, Industry Type dropdowns
- **Step 3 (Custom only) - Columns:** Checkbox selection of which data columns to include
- **Step 4 - Format:** PDF (branded), Excel (.xlsx), or CSV (raw)
- **"Preview Data" button:** Shows first 5 rows in a table
- **"Generate Report" button:** Creates and downloads the report file
- **"Save Configuration" button:** Saves current filters to localStorage for reuse

- **"My Saved Reports" button:** Opens dialog to load previously saved configurations
- **Recent Reports Table:** Previously generated reports with download buttons

Settings /settings

System configuration:

- **Security Section:** Toggle 2FA, password policy, IP restriction switches
- **Data Backup:** Toggle auto-backup, set frequency, trigger manual backup
- **System Info Sidebar:** Version, uptime, last backup, database status

5. Government Officer Guide

Login: govt@tn.gov.in

Sidebar Navigation

COMMAND & CONTROL

- Command Center - State-wide KPIs and oversight
- Parks Explorer - Interactive park information
- Services Overview - Track all service requests

MONITORING

- Compliance Engine - Violation monitoring
- Analytics - Investment & employment analysis

REPORTS

- Report Center - Generate official reports

Primary Workflow: Daily Monitoring

1 Login and review Command Center

Check the 5 KPI cards for any significant changes. Pay attention to the Red Flags count.

2 Review Alerts

Check the Alerts panel on the Command Center. Critical alerts (red) need immediate attention.

3 **Check Compliance Engine**

Navigate to Compliance Engine to see open violations, missing submissions, and send reminders.

4 **Monitor Services**

Check Services Overview for overdue requests and bottlenecks.

5 **Generate Reports as needed**

Use Report Center for official documentation, investor presentations, or policy reviews.

Key Differences from Admin

Feature	Govt Officer	Admin
Command Center	Full access	Full access
User Management	No access	Full CRUD
Compliance Engine	View only	View + Edit violations
Services Tracker	View + Update status	Full control
Settings	No access	Full access
Report Center	Full access	Full access

6. Industry User Guide

Login: industry@abc.com

Sidebar Navigation

MY WORKSPACE

- Workspace - Personal dashboard with compliance score
- Company Profile - Edit company details

COMPLIANCE

- Submit Data - Quarterly/annual data submission
- My Compliance - Compliance status view
- Service Requests - NOC and allotment tracking

REPORTS

- Report Center - Download own reports
- Parks Explorer - View all industrial parks

Industry Workspace /workspace

Your personal dashboard - the first screen after login:

- **Company Header:** Shows company name, park, plot number, industry type with gradient banner
- **Compliance Health Score:** Circular score gauge (0-100) with breakdown bars:
 - Submission Score (90/100)

- Environmental Score (72/100)
- Financial Score (85/100)
- Safety Score (65/100)
- **Actionable Tasks:** Checklist of pending items with priority levels (high=red, medium=orange). Completed items shown with strikethrough
- **Compliance Notices:** Yellow warnings and blue info notices
- **Quick Stats:** Employees, Investment, Power Usage, Water Usage in 4 mini cards
- **Lease Status:** Shows validity dates and monthly rent. Click "View Lease Agreement" for full details dialog with download option
- **Document Vault:** Table of uploaded documents with category, file name, expiry date, and verification status. Click "Upload Document" to add new files
- **Service Requests:** Mini status board showing Applied/In Review/Approved/Completed counts

Submit Data (Unified Data Submission) </submit-data>

Two submission modes available:

Mode 1: Step-by-Step Form

- 1 **Period:** Select Year (2025/2026) and Quarter (Q1-Q4)
- 2 **Financial:** Investment Amount, Annual Turnover, Export Revenue, R&D Expenditure (all in Crores)
- 3 **Employment:** Permanent, Contract, SC/ST, and Women employee counts
- 4 **Resources:** Water Consumption (KL), Power Usage (kWh), Waste Generated (Tons), Waste Recycled (%)
- 5 **CSR:** Activities description, Amount Spent (Lakhs), Beneficiaries count
- 6 **Review & Submit:** See all data in summary cards. Click "Submit Data" to finalize or "Save Draft" to continue later

Mode 2: Excel Upload

- 1 Click "Download Template (.csv)" to get the pre-formatted template file

- 2 Fill in data in the template using Excel or any spreadsheet software
- 3 Click "Choose File" and select your filled template
- 4 Click "Upload & Validate" - the system checks all fields and reports any errors

Submission History: Table below the form shows all past submissions with status (Approved/Draft/Submitted) and action buttons (View/Edit).

Service Requests </services>

Track NOC applications, allotments, and connections:

- **" + New Request" button:** Opens dialog to create a service request:
 - Select Service Type (NOC Fire, Land Allotment, Water/Power Connection, etc.)
 - Set Priority (Normal/High/Urgent)
 - Add Remarks/Description
 - Click "Submit Request" - get confirmation with reference number
- **Kanban View:** See your requests flow through stages
- **Click any request:** View milestone timeline with completion dates
- **"Withdraw Request":** Cancel a pending request with confirmation

Company Profile </industry-profile>

Manage your company details:

- View and edit Contact Person name and Phone number
- Upload GST Certificate file
- Click "Save Changes" to persist updates

7. Public Pages (No Login Required)

Page	URL	Description
Home	/home	Landing page with industrial pulse stats, feature showcase, park highlights, and CTA to login
The Initiative	/about	About SIMS - vision, mission, security commitments, economic impact
Capabilities	/features	Feature showcase - Surveillance, Compliance, Insights, Verification
Industrial Parks	/parks	Featured parks with images - Oragadam, Sriperumbudur, Hosur
Support	/contact	Contact form with helpline, email, and office address. Submit form for 4-6hr response

Home Page Navigation

The top navigation bar includes:

- **SIPCOT SIMS logo:** Always visible
- **The Initiative:** Links to About page
- **Capabilities:** Links to Features page
- **Industrial Parks:** Links to Parks page
- **Support:** Links to Contact page
- **Portal Login:** Navigates to Role Selection page

On mobile, the hamburger menu icon navigates directly to the role selection page.

Footer Links

All footer links are functional: Platform links navigate to their respective pages, Resources links navigate to Features/About/Contact, and Privacy Policy/Terms link to the About page.

8. Shared Features

Notification System

The bell icon in the top-right of the AppBar shows a badge with the count of new notifications. Click to expand the notification panel with recent system alerts.

Parks Explorer (All Roles)

Available to all authenticated users at `/parks-explorer`. Shows 3 featured parks with images, statistics, and location info. "View All 21 Locations" and "Open GIS Portal" buttons redirect to login for full access.

Report Center (All Roles)

The Report Center is accessible to all authenticated users, but with role-based data access:

- **Admin/Govt:** Can generate reports across all industries and parks
- **Industry:** Can only generate reports for their own data

Logout

Click the "Logout" button at the bottom of the sidebar. This clears your authentication token and redirects to the login/role selection page.

Responsive Design

The application is fully responsive. On mobile devices:

- The sidebar becomes a slide-out drawer (tap the hamburger menu icon)
- Grid layouts stack vertically
- Charts resize to fit the screen

9. Frequently Asked Questions

Q: I logged in but see a different dashboard than expected?

A: Make sure you selected the correct portal on the Role Selection page. If you're an Industry user but selected the Admin portal, the system will show an "Access Denied" error. Go back to /role-selection and choose the correct portal.

Q: How do I submit quarterly data?

A: Login as Industry user > Click "Submit Data" in sidebar > Select period > Fill in all 5 steps > Click "Submit Data" on the Review page. You can also save as draft and continue later.

Q: Can I download my submission data?

A: Yes, go to Report Center > Select the report type > Choose your period and filters > Click "Generate Report" to download as PDF, Excel, or CSV.

Q: How do I know if I'm compliant?

A: As an Industry user, your Workspace shows a Compliance Health Score (0-100) with breakdown by category. Keep all scores above 80 for "Excellent" status.

Q: How do I apply for a new NOC or service?

A: Go to Service Requests > Click "+ New Request" > Select service type, priority, add description > Submit. You'll receive a reference number and can track progress on the Kanban board.

Q: The data shown is mock/sample data. Will it change?

A: Yes, the current system uses mock data for demonstration. Once the PostgreSQL database is connected and migrated (using schema.sql + schema_v2.sql), all data will be live from the database.

Q: How to print this guide as PDF?

A: Open this HTML file in any browser, then use **Ctrl+P** (or Cmd+P on Mac) and select "Save as PDF" as the destination.

10. Technical Support

Contact Information

Channel	Details
Toll-Free Helpline	1800-425-XXXX (24/7)
Technical Support Email	support-sims@sipcot.tn.gov.in
Industrial Liaison Email	industry-help@sipcot.tn.gov.in
Head Office	SIPCOT Projects Office, 19-A, Rukmani Lakshmipathy Salai, Egmore, Chennai - 600008
Response Time	4-6 business hours
Maintenance Window	Sunday 02:00 AM - 04:00 AM IST

Starting the Application (Development)

Start Backend:

```
cd backend
npm install
npm run dev
```

Start Frontend (new terminal):

```
cd frontend
npm install
```

```
npm run dev
```

Access:

Frontend: <http://localhost:5173>

Backend API: <http://localhost:5001>

Quick Start: After starting both servers, open <http://localhost:5173> in your browser. You'll see the SIPCOT SIMS home page. Click "Portal Login" to begin.

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